# JIXIAN LIU

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## APPOINTMENTS

**Graduate Research Assistant** Aug. 2025 - Present
NetD-Lab, Data Science and AI Institute, Advisor: Enrique Mallada
Johns Hopkins University

## RESEARCH INTERESTS

Robotics, Robot Learning, Safety-Critical Control, Reinforcement Learning.

## EDUCATION

**Johns Hopkins University** Baltimore, MD Aug. 2024 - Present
Ph.D. Electrical and Computer Engineering

**Xi'an Jiaotong University** Xi'an, Shaanxi Sept. 2020 - Jul. 2024
B.S. Physics

## HONORS & AWARDS

- CoRL Travel Award 2026
- Amazon AI PhD Fellowship ($155,000) 2026
- ACC Travel Grant 2026
- Mathematical Institute for Data Science Fellowship ($25,000) 2026
- Hong Kong PhD Fellowship (Top Award for HK PhD Applicants) 2024
- Dean's Fellow Award, Boston University (Top Award for BU PhD Applicants) 2024
- MITACS Globalink Research Internship Scholarship (C$7000) 2023
- Top-10 Undergraduate Students of the Year (Top Individual Award in XJTU) 2023
- National Scholarship (Top 1%) 2022, 2023
- Finalist Winner, Mathematical Contest in Modeling (Top 2% around the World) 2022
- First Place in Shaanxi, The Chinese Mathematics Competitions for College Students 2021
- Presidential Scholarship, Xi'an Jiaotong University (Top 5‰) 2021

## RESEARCH PUBLICATIONS

### Conferences & Workshops

1. **Jixian Liu**, Ihab Tabbara, Hussein Sibai and Enrique Mallada, "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity", 10th Annual Conference on Robot Learning, 2026.
2. Zhuo Ouyang, **Jixian Liu**, and Enrique Mallada, "Symplectic Inductive Bias for Data-Driven Target Reachability in Hamiltonian Systems", 65th IEEE Conference on Decision and Control, 2026.
3. **Jixian Liu** and Enrique Mallada, "Safety-Critical Control via Recurrent Tracking Functions", 2026 American Control Conference, 2026.
4. **Jixian Liu** and Enrique Mallada, "Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification", 64th IEEE Conference on Decision and Control, 2025.

5. **Jixian Liu**, Tao Xu, Jianping He, and Chongrong Fang, "Smart Predict-then-Optimize Method with Dependent Data: Risk Bounds and Calibration of Autoregression", arXiv preprint arXiv:2411.12653, 2024.

**Journals**

1. Xueyuan Cui, **Jixian Liu**, and Yao Zhou, "Enabling Net Zero Energy Buildings With Shared Storage: A Cyber-Physical Perspective", IEEE Internet of Things Journal, 2023.
2. **Jixian Liu**, Ihab Tabbara, Hussein Sibai and Enrique Mallada, "Fundamental Limits of Sampling-Based Reachability: Geometry and Flow Distortion", arXiv preprint arXiv:2610.40473, 2026.

**Datasets & Benchmarks**

1. Yiyou Sun, Xinyang Han, Weichen Zhang, ..., **Jixian Liu**, Maojia Song, Ruheng Wang, Shengcao Cao, Yanxin Zhang, et al., "Agents' Last Exam," arXiv preprint arXiv:2606.05405, 2026.

## ACADEMIC SERVICES

**Reviewer of Journals**

- IET Renewable Power Generation Journal
- IEEE Control Systems Letters

**Reviewer of Conferences & Workshops**

- American Control Conference 2026, 2027
- IEEE Conference on Decision and Control 2026
- China Automation Congress 2024

## TEACHING

**Teaching Assistant**

- Foundation of Reinforcement Learning (Online), Johns Hopkins University Fall 2025, Fall 2026
- Control System Design, Johns Hopkins University Fall 2025

## ADVISING AND MENTORING

**Mentoring**

**Zhuo Ouyang** Undergrad Student, Peking University

College of Engineering Next position: CS Ph.D. at UIUC (Advisors: Tong Zhang and Nan Jiang)

## PRESENTATIONS

**Talks**

- Oct. 2026, "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity," Invited talk, hosted by Lars Lindemann, Zurich, Switzerland
- May 2026, "Safety-Critical Control via Recurrent Tracking Functions", American Control Conference, New Orleans, LA, USA
- Dec. 2025, "Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification", 64th IEEE Conference on Decision and Control, Rio de Janeiro, RJ, Brazil
- Oct. 2025, "Recurrence is All You Need: From Parametric to Nonparametric Safety", Institute for Assured Autonomy Graduate Student Lightning Talks, Baltimore, MD, USA

**Poster**

- Nov 2026 "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity", 10th Conference on Robot Learning, Austin, TX, USA

- Oct 2026, "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity", JHU & Amazon AI2AI Research Festival, Baltimore, MD, USA
- May 2025, "Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification", 2nd Northeast Systems and Control Workshop, NYC, USA

## REFERENCES

**Enrique Mallada** Ph.D. Advisor
Associate Professor, Department of Electrical and Computer Engineering
Johns Hopkins University, Baltimore, MD

**Jianping He**
Professor, School of Automation and Intelligent Sensing
Shanghai Jiao Tong University, Shanghai, SH

**Xiaoming Duan**
Associate Professor, School of Automation and Intelligent Sensing
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